are used in this thesis. First, the different probability distributions are determined and discretized and finally multiplied together. This has the disadvantage of some inaccuracy, but the advantage of less computational cost. Second, Markov chain Monte Carlo (MCMC) methods are used. In this process, the various probability distributions are sampled such that the Markov chain converges to the correct posterior distribution. In Sect. 2.4, two MCMC methods are described in more detail, namely the Gibbs and Metropolis-Hastings sampling methods (Betancourt, 2017).

The flowchart in Fig. 2.11 shows how the basic principles of Bayesian statistics just described are used in the following sections. This includes Bayesian hierarchical modeling with respect to age-depth/distance transformations solved with matrix multiplications. The purpose of this step is to allow different climate proxies to be synchronized with respect to age. In parallel, transfer functions are inversely determined using various machine learning techniques to establish a probabilistic relationship between plants and climate. The next step is a BHM of local climate reconstructions based on the data just described and solved with MCMC. All this leads to the possibility of spatial climate reconstructions, also based on a BHM and solved using MCMC.

2.3 Age-depth/distance modelling

This section deals with the relationship between age dating and depths in lake and mire sediments and distances in speleothems. The various methods of age dating can be found in Bradley, 2014. For the relationship between age and depth/distance, we use a Bayesian age-depth/distance model in order to account for age uncertainties. In contrast, most previous studies are based solely on the mean values. For example, depths of lake sediments (Litt et al., 2012; Schiebel and Litt, 2018; Torfstein et al., 2015; Neumann et al., 2007; Miebach et al., 2019; Seppä et al., 2005) or distances (from a reference point) with respect to speleothems (Drăguşin et al., 2014; Waltgenbach et al., 2020; Akers et al., 2016; Psomiadis et al., 2018). For the latter, the values are based on models such as StalAge (Scholz and Hoffmann, 2011) and Bacon (Blaauw and Christen, 2011). In our work, we instead consider the entire uncertainty domain. The Bayesian age-depth model Bacon just mentioned, which is implemented in R, is used for this purpose (R Core Team, 2018; Blaauw et al., 2020). We will only briefly explain the model in this thesis. A detailed description can be found in Blaauw and Christen, 2011.

The Bacon model uses a self-adjusting MCMC simulation to infer the posterior probability distribution $\mathbb{P}(\theta, x, \omega|y)$, where θ is the model parameter, x is the accumulation rate, ω is the memory, and y are measurements such as ^{14}C data. These are the same variable declarations as in Blaauw and Christen, 2011 and apply only to this section. The middle and right panels in the top row of Fig. 2.12 show an example of the prior distributions of x and ω, which are indicated as green lines. Based on the MCMC samples, the gray areas represent the corresponding posterior probability densities. From such a posterior accumulation rate x, we can obtain the probability distributions $\mathbb{P}(\boldsymbol{A}|\boldsymbol{D})$ and $\mathbb{P}(\boldsymbol{D}|\boldsymbol{A})$. $\boldsymbol{A}$ is the age and $\boldsymbol{D}$ denotes the depth or distance. The Bacon.Age.d function in the Bacon

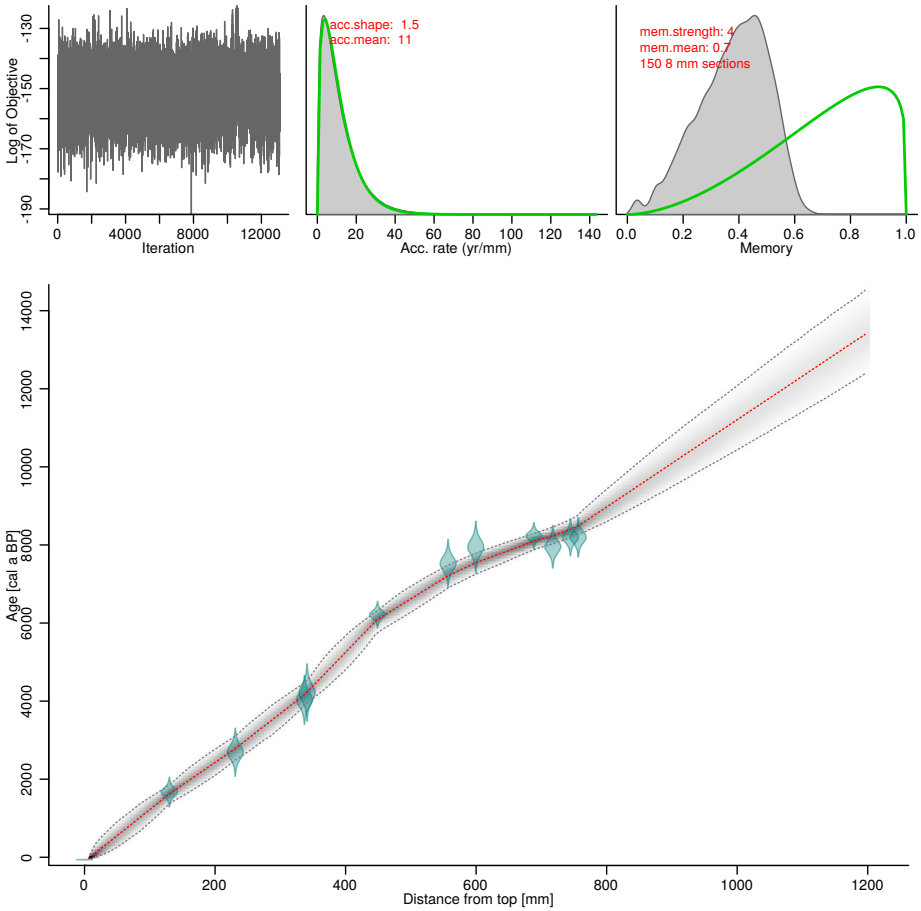


Figure 2.12: The lower panel shows an age-distance model of a speleothem from the Ascunsă Cave in Romania calculated with Bacon. The top panels depict, from left to right, a trace plot of the posterior MCMC samples, prior (green lines) and posterior (gray area) distributions of accumulation rate and memory.

package returns $\mathbb{P}(\boldsymbol{A}|\boldsymbol{D})$. In our study, however, we need a tool for $\mathbb{P}(\boldsymbol{D}|\boldsymbol{A})$ (see Eq. 2.10) which has been implemented since package version 2.5 and can be found under Bacon.d.Age. Examples of both probability densities are shown in Fig. 2.13, namely $\mathbb{P}(\boldsymbol{A}|440\,\text{mm})$ and $\mathbb{P}(\boldsymbol{D}|6000\,\text{cal a BP})$. The corresponding age-distance model can be seen in Fig. 2.12. It is based on the dating of a speleothem from the Ascunsă Cave in Romania (Drăguşin et al., 2014). Originally, the Bacon model was created for the calculation of age-depth models based on ^{14}C dating from cores of, e.g., peat bogs or lake sediments. That is, Bacon performs an appropriate calibration from ^{14}C a BP to cal a BP. However, this option can be disabled and the model assumes that the calibrated dates are already available, which is the case here. Such datings and their uncertainties can be taken from the speleothem database SISAL. Roesch and Rehfeld (2019) tested 88 entities for robustness to various age-distance models. Among them are Bchron (Haslett and Parnell, 2008) and linear interpolations as well as Bacon and StalAge. Linear interpolations, Bchron and Bacon provide robust results for these entities and the

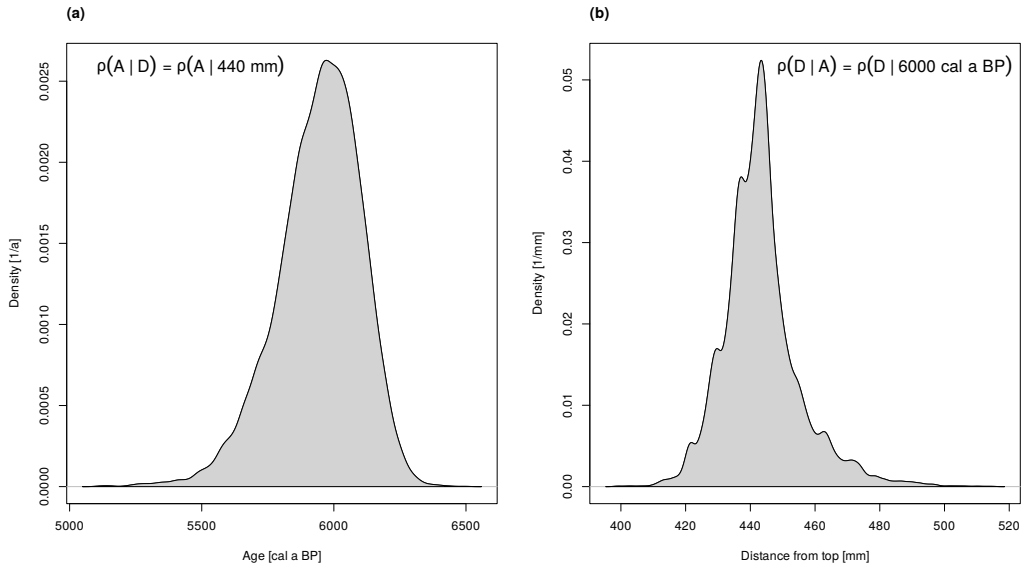


Figure 2.13: Probability density distributions based on the age-distance model in Fig. 2.12. Panel (a) depicts $\mathbb{P}(\boldsymbol{A}|440\,\text{mm})$ calculated with the R function Bacon.Age.d and (b) $\mathbb{P}(\boldsymbol{D}|6000\,\text{cal a BP})$ calculated with the R function Bacon.d.Age.

latter is the only one that can account for hiatus. Therefore, Bacon is also suitable as an age-distance model for speleothems. In the upper middle panel of Fig. 2.12, some prior parameters defining the gamma-distributed accumulation rate are also shown. These values of x are set to a mean accumulation rate of 11 mm/year and a of shape 1.5. This mean is calculated from the assumption of a uniform slope over the entire age-distance range, and the shape is the default value, which provides a flatter curve and thus more variability. As the growth rate of this speleothem is relatively constant (except the 8.2 event) during the Holocene (Drăguşin et al., 2014), the default settings of the memory are chosen, namely mean memory of 0.7 and memory strength of 4 (upper right panel). Overall, the trace plot of $\log(\mathbb{P}(\theta, x, \omega|y))$ in the upper left panel shows good mixing, indicating stable model performance with these prior parameters.

Now we need to determine how to use the information from this model to make an appropriate conversion from distances to ages. As mentioned above, the probability distributions $\mathbb{P}(\boldsymbol{D}|\boldsymbol{A})$ are crucial for this, which is shown by the following derivation using the technique of Bayesian Hierarchical Models:

$$\mathbb{P}(\boldsymbol{Y}, \boldsymbol{A}, \boldsymbol{P}) = \int_{\mathcal{D}} \mathbb{P}(\boldsymbol{Y}, \boldsymbol{A}, \boldsymbol{P}, \boldsymbol{D})\, d\,\boldsymbol{D} \tag{2.8}$$

$$= \int_{\mathcal{D}} \mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{A}, \boldsymbol{P}, \boldsymbol{D}) \cdot \mathbb{P}(\boldsymbol{D} \mid \boldsymbol{A}, \boldsymbol{P}) \cdot \mathbb{P}(\boldsymbol{A}, \boldsymbol{P})\, d\,\boldsymbol{D}. \tag{2.9}$$

$\boldsymbol{Y}$ contains variables such as annual precipitation or δ^{18}O for which an age-depth/distance transformation

is to be performed. $\boldsymbol{P}$ means proxy information such as pollen, macrofossils, or any type of isotopic data. In the following, we assume that $\boldsymbol{D}$ is conditionally independent of $\boldsymbol{P}$ and thus completely dependent on $\boldsymbol{A}$. This is exactly the information we get from the age-depth/distance model. Moreover, the variables $\boldsymbol{Y}$ should not depend on age given $\boldsymbol{D}$. This assumption stems from the fact that all the proxy information we consider initially refers to a depth or distance. Then, we obtain the final age-depth/distance transformation:

$$\Rightarrow \mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{A}, \boldsymbol{P}) = \int_{\mathcal{D}} \mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{D}, \boldsymbol{P}) \cdot \mathbb{P}(\boldsymbol{D} \mid \boldsymbol{A})\, d\,\boldsymbol{D}. \tag{2.10}$$

In this thesis, the conditional probability distributions are discretized and stored and evaluated in predefined matrices.

Examples of a transformation using Eq. 2.10 can be seen in Fig. 2.14. Blue represents δ^{18}O, red represents δ^{13}C, and black represents the respective smoothings of these values. Each of these 6 different curves can be understood as Y, except that their calculations are different. The colored curves are based on the mean values of the Bacon model from Fig. 2.12 (dashed red line). The dashed black curves are analogous to those in Drăguşin et al. (2014) using a 9-point moving average. In contrast, the solid black lines are calculated with the use of Eq. 2.10.

Here, $\mathbb{P}(\boldsymbol{Y}|\boldsymbol{D}, \boldsymbol{P})$ represents the relationship of δ^{18}O and δ^{13}C to their distance (with respect to the

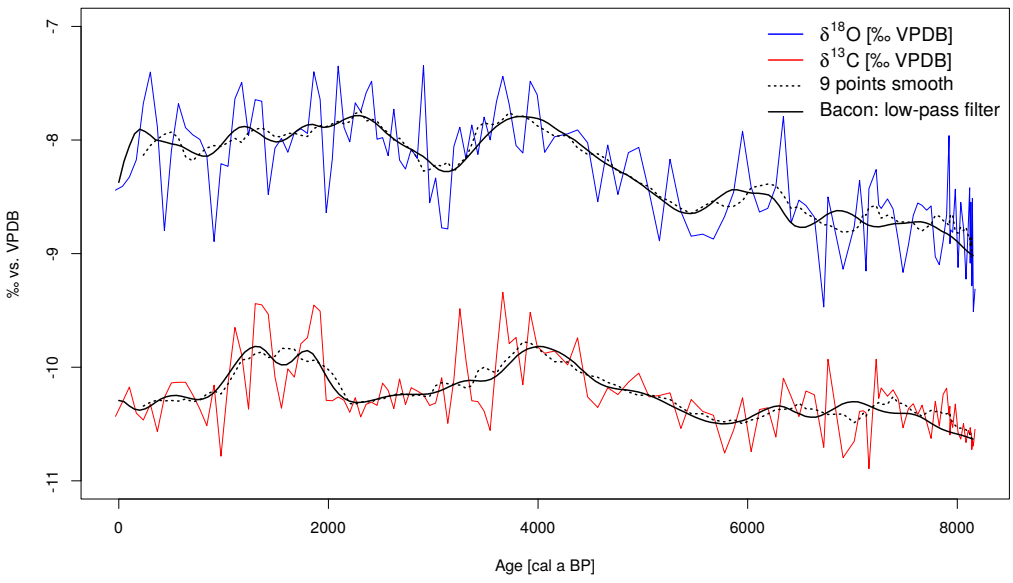


Figure 2.14: δ^{18}O (blue line) and δ^{13}C (red line) data from Drăguşin et al., 2014. The dashed black lines represent their moving average, and the solid black lines show our data-driven smoothing using the probabilistic relationship between age and distance.

top of the speleothem) and to $\boldsymbol{P}$, which can be interpreted as the total isotopic information of ^{18}O and ^{13}C, respectively. Thus, the blue and red curves can be defined as $\mathbb{P}(\delta^{18}\mathrm{O}|\boldsymbol{D},\boldsymbol{P})$ and $\mathbb{P}(\delta^{13}\mathrm{C}|\boldsymbol{D},\boldsymbol{P})$. The temporal resolution of $\mathbb{P}(\boldsymbol{D}|\boldsymbol{A})$ can be chosen arbitrarily high. In this thesis, this value is set at 50 years, which is a compromise between all speleothems and sediment cores. If it is specified evenly, this transformation results in a regular time grid. A regular grid is one requirement for the application of the wavelet transform introduced above. Nevertheless, there must be an age-depth/distance relationship per entity. Accordingly, such transformation cannot be performed for entities such as LR04, Medstack, and NGRIP (see. Fig. 2.2). Mathematically, Eq. 2.10 behaves like a convolution with different kernel smoother (cf. Eq. 2.2). For this reason, a smoothed pattern is obtained when age uncertainties are taken into account. Unlike traditional smoothing methods, this one is data-driven and shows what

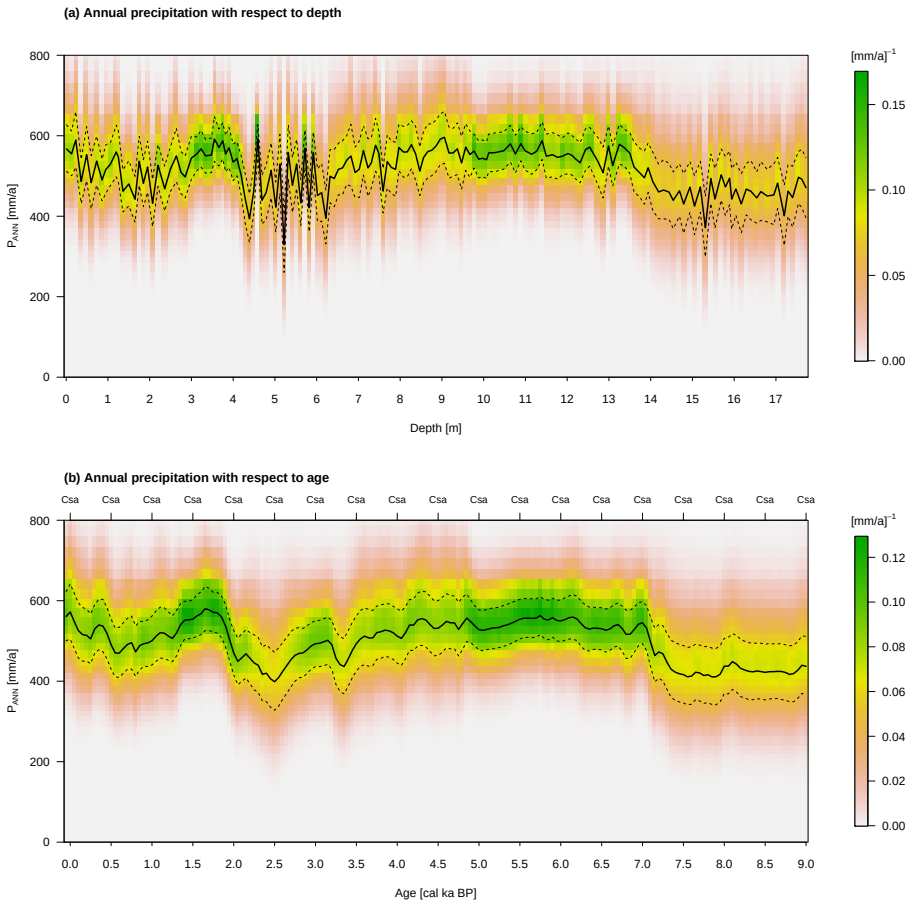


Figure 2.15: Panel (a) shows the climate reconstruction of annual precipitation as a function of the depth of the lake sediment of the Sea of Galilee. The colored contours represent the uncertainties in terms of probability densities, and the dashed black lines depict the interquartile range based on these data. In it, the median is shown in the form of a solid black line. Panel (b) corresponds to (a) with the difference that the x-axis is transformed from depth to age.

information $\boldsymbol{Y}$ actually provides in terms of age.

Another advantage of this method is its simple implementation and low computational cost. Previously, age uncertainties were based on sampling algorithms as in Parnell et al. (2014) or on ensembles of age depth/distance models (McKay et al., 2020). In our case, we only need to create $\mathbb{P}(\boldsymbol{D}|\boldsymbol{A})$ and store it in a matrix with dimensions $m \times n$. m is the number of points with respect to $\boldsymbol{D}$ and n is the user-dependent number of data points related to age. Pragmatically, $\mathbb{P}(\boldsymbol{Y}|\boldsymbol{D}, \boldsymbol{P})$ has dimension $1 \times m$ in the case of Fig. 2.14. Thus, Eq. 2.10 can be calculated simply by matrix multiplication (denoted here by $\otimes$):

$$\mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{A}, \boldsymbol{P}) \in^{1\times n} = \mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{D}, \boldsymbol{P}) \in^{1\times m} \otimes \mathbb{P}(\boldsymbol{D}|\boldsymbol{A}) \in^{m\times n} .$$

This fact is taken up again in Sect. 2.4, where the advantage of the low computational cost of this multiplication is exploited.

$\mathbb{P}(\boldsymbol{Y} \mid \boldsymbol{D}, \boldsymbol{P})$ can also have the dimension $k \times m$. $k > 1$ denotes the number of grid points with respect to $\boldsymbol{Y}$. In Fig. 2.15, $\boldsymbol{Y}$ is the mean annual precipitation ($\mathrm{P_{ANN}}$). Probability density values are available for this variable at each depth (colored contour in (a)). The dimension for the discretized probability density function (PDF) is also user-dependent and is set to $k = 300$ in this thesis. The solid black line represents the median of each PDF. All in all, the transformation from depth (a) to age (b) follows the same procedure as the one described above. In total, the following is calculated:

$$\mathbb{P}(\mathrm{P_{ANN}} \mid \boldsymbol{A}, \boldsymbol{P}) \in^{k\times n} = \mathbb{P}(\mathrm{P_{ANN}} \mid \boldsymbol{D}, \boldsymbol{P}) \in^{k\times m} \otimes \mathbb{P}(\boldsymbol{D}|\boldsymbol{A}) \in^{m\times n} . \tag{2.11}$$

This example serves only to illustrate the transformation from depth (a) to age (b). A corresponding description from a paleoclimatological perspective can be found in Sect. 3.3.2. How such a reconstruction is calculated in detail is described in the following.

2.4 Local climate reconstructions

The climate reconstruction model used in this thesis is based, among other things, on the idea of the indicator species approach. In this method, pollen or macrofossils from deposits, such as lake and mire sediments, are considered as climate indicators. Using this approach, Iversen, 1944 reconstructed January and July temperatures during the Holocene in Scandinavia based on *Ilex*, *Viscum*, and *Hedera*. Building on these basics, the Mutual Climate Range (MCR) method has evolved (Grichuk, 1969). Here, plant distribution maps are linked to recent climate data to define a climate range where the corresponding taxon occurs. Finally, when considering multiple taxa, these climatic ranges can be combined to determine the mutual climatic range. This method brings a number of challenges. First, there is no weighting within the individual climate ranges. This means, for example, that each temperature range occurs with the same probability. These sharp boundaries in transfer functions

result from overfitting (Friedman, 2001). This is a general problem in determining such functions and will be taken up again in more detail in the next section. On the other hand, combining too many taxa can result in more than one MCR. In contrast, when too few taxa occur in a certain sediment layer, reconstruction can lead to a relatively broad climatic range (Klotz et al., 2004). In addition, only the occurrence of the plants is considered. In this method, the percentage information is included only indirectly, by determining a priori whether a plant is present or not. There are some other climate reconstruction methods that incorporate such percentage distributions. These include regression methods in which linear or nonlinear transfer functions are calculated based on modern climate and plant data (Webb and Bryson, 1972). In addition, there are classification methods, such as Modern Analogue Technique (MAT) (Overpeck et al., 1985). The same machine learning (ML) algorithms can be used in both regression and classification (Jed Wing et al., 2019). These are among others Artificial Neural Network (ANN) or Random Forest (RF) (see Sect. 2.5). Chevalier et al., 2020 present a table describing the advantages and disadvantages of the above methods. One advantage of the MCR method is that it can be used globally. In contrast, the regression and classification approaches are more spatially constrained and often have poorer estimates of uncertainty. In addition to these methods, Chevalier et al., 2020 mention two more: Bayesian approaches and inverse modeling. Both are also used in this thesis, so that an attempt is made to balance the disadvantages of all methods with their respective advantages. In the following, we describe how we incorporate a Bayesian approach into our reconstruction model.

To do this, we follow the idea of Kühl et al., 2002, who developed a probabilistic interpretation of MCR. Fig. 2.16 shows the concept of this method. This addresses the problem of overfitting by calculating uncertainty ranges for each taxon. These are shown here for taxa A, B, and C for different colored contours (blue, brown, and green). The MCR follows the colors of the color bar. The uncertainty ranges were initially based on normally distributed PDFs (with respect to temperature), which is why this is called the PDF method. Later, more complicated models were also used, which will be discussed in more detail in Sect. 2.5. The PDF method was applied for both local and spatial climate reconstructions. For example, Kühl and Litt, 2003 calculated January and July temperatures for 3 sites in Central Europe during the Last Interglacial (LIG) period. Subsequently, spatial reconstructions of Europe were performed in Gebhardt et al., 2008 for the Eemian, in Simonis et al., 2012 for the Late Glacial and Holocene, and in Weitzel et al., 2019 for the MH. Schölzel, 2006 describes the PDF method in the context of a BHM and calls it Bayesian Indicator Taxa Model (BITM). This has the advantage that additional prior information can regulate the transfer functions. Among others, the BITM was applied in Neumann et al., 2007 for Birkat Ram in Israel, in Stolzenberger, 2011 for Meerfelder Maar and Holzmaar in Germany, and in Thoma, 2017 for Lake Prespa in Greece. In all BITM-based reconstructions, thresholds must be set to define at which point a taxon occurs or does not occur. Another method presented by Schölzel, 2006 that is based on BHM is the so-called Bayesian Biome Model (BBM). In this process, certain plant taxa are assigned to different biomes. These are

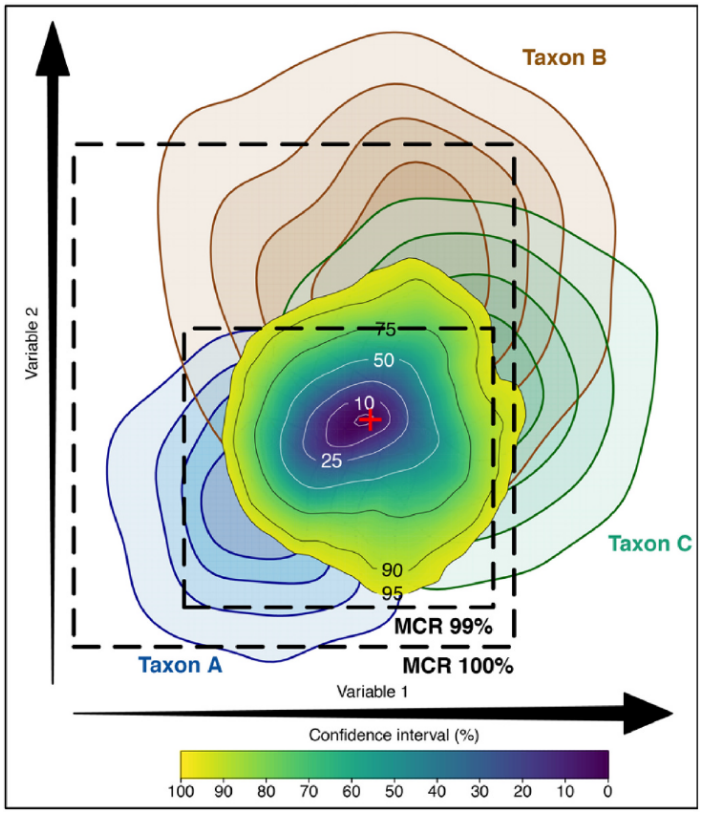


Figure 2.16: Illustration of a Mutual Climate Range from Chevalier et al., 2020. Shown are 3 taxa as a function of two climate variables and their MCR, represented by the yellow/blue contours. The red cross represents the mean of the MCR, and the dashed rectangles demonstrate the sensitivity of this technique when only a subset of the MCR data are considered. That is, if the input data contain 99 %, there may be a relatively large difference from the full input data set (MCR 100 %).

groups that have similar climatic conditions and predominant vegetation, among other things (Prentice et al., 1992). One advantage of the BBM is that one does not need recent distribution maps for every plant occurring in the core, but only for the biomes used. The computational assignment of taxa to biomes is based on the affinity score (Prentice et al., 1996). This is used to calculate the percentage distributions of the biomes with respect to the depth of the core. Nevertheless, the information of a threshold value of the individual taxa is included in the affinity score. Applications of the BBM can be found in Litt et al., 2012, in Stolzenberger, 2017, and Litt et al., 2021 for the Dead Sea in Israel. Based on the above summary, the first part of the reconstruction method used here is now described. The inverse modeling is done in the second part, which is discussed below. The goal of the new method is to provide an algorithm that performs the calculations regardless of location across Europe. This is intended to circumvent expert knowledge for setting thresholds for taxa occurrence, which is a major effort for the approximately 600 sites. Instead, the new method should independently determine which taxa are important for each reconstruction based on specific constraints. All variables that are relevant

Table 2.2: List of variables.

Variable shortcut	Description
$\mathbb{P}$	Probability distribution
$\boldsymbol{C}$	Climate: contains modern $\boldsymbol{C_m}$ and past $\boldsymbol{C_p}$ climate information
$\boldsymbol{P}$	Proxy: contains modern $\boldsymbol{P_m}$ and past $\boldsymbol{P_p}$ proxy information
$\boldsymbol{PP}$	Proxy pool: explained variances between additional proxies and $\boldsymbol{C}$
$\boldsymbol{P_s}$	Selected plant proxy information
$\boldsymbol{A}$	Age of sediments or speleothems
$\boldsymbol{D}$	Depth of sediments or distance in speleothems
$\boldsymbol{\Theta}$	Contains the following parameters:
$\boldsymbol{\psi}$	Link between climate $\boldsymbol{C}$ and taxa distributions $\boldsymbol{P}$
$\boldsymbol{\phi}$	Contains all information about the taxa weights:
$\boldsymbol{\omega}$	Taxa weights
$\boldsymbol{z}$	Selection of taxa based on taxa weights $\boldsymbol{\omega}$

for the mathematical formulation of this model can be found in Table 2.2.

Using Bayes' theorem, we can express the probability distribution for climate $\boldsymbol{C}$ given pollen and macrofossils $\boldsymbol{P}$, depth $\boldsymbol{D}$, and parameter $\boldsymbol{\Theta}$. In the process, we also introduce the selected plant proxies $\boldsymbol{P_s}$:

$$\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\Theta}) = \int_{\mathcal{P}_s} \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{P_s}, \boldsymbol{D}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\Theta})\, d\boldsymbol{P_s}. \tag{2.12}$$

In the case of a finite number of taxa, the integral is a corresponding sum. Consider $\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{P_s}, \boldsymbol{D}, \boldsymbol{\Theta})$ in more detail:

$$\begin{aligned}\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{P_s}, \boldsymbol{D}, \boldsymbol{\Theta}) &\overset{1.}{\approx} \mathbb{P}(\boldsymbol{C} \mid, \boldsymbol{P_s}, \boldsymbol{D}, \boldsymbol{\Theta}) \overset{2.}{\approx} \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P_s}, \boldsymbol{\Theta}) \overset{3.}{\approx} \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P_s}, \boldsymbol{\psi}) \\ &\overset{4.}{\approx} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})} \\ &\overset{5.}{\approx} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})}.\end{aligned} \tag{2.13}$$

With the following assumptions and applications:

1. $\boldsymbol{C}$ is conditionally independent of $\boldsymbol{P}$ if $\boldsymbol{P_s}$ is given. This assumes that $\boldsymbol{P_s}$ explains enough variability of the core.

2. The link between $\boldsymbol{C}$ and $\boldsymbol{P_s}$ is conditionally independent of depth. This means that the relationship between these quantities is assumed to be unchanged for any depth and thus any age of the core. The assumption that this relationship has not changed over time is an important part of our reconstruction method. When we look at older time periods, we need to keep this in

mind, as the relationship may well have changed due to evolutionary processes.

3. The connection between $\boldsymbol{C}$ and $\boldsymbol{P_s}$ is described only by the parameter $\boldsymbol{\psi}$. Furthermore, $\boldsymbol{\psi}$ and $\boldsymbol{\phi}$ are a priori independent: $\mathbb{P}(\boldsymbol{\Theta}) = \mathbb{P}(\boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\phi})$.

4. Application of Bayes' theorem.

5. If $\boldsymbol{C}$ depends only conditionally on $\boldsymbol{\psi}$, it is assumed that within the considered climate ranges each value occurs with the same probability. Thus, $\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{\psi})$ follows a uniform distribution.

If we now substitute Eq. 2.13 into equation Eq. 2.12, we get:

$$\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\Theta}) \approx \int_{\mathcal{P}_s} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})} \cdot \mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\phi})\, d\boldsymbol{P_s}. \tag{2.14}$$

Combined with Eq. 2.10, the following expressions can be found:

$$\begin{aligned}
\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) &= \int_{\mathcal{D}} \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{D} \mid \boldsymbol{A})\, d\boldsymbol{D} \\
&\overset{\text{Eq.2.14}}{\approx} \int_{\mathcal{D}} \int_{\mathcal{P}_s} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})} \cdot \mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\phi})\, d\boldsymbol{P_s} \cdot \mathbb{P}(\boldsymbol{D} \mid \boldsymbol{A})\, d\boldsymbol{D} \\
&\approx \int_{\mathcal{P}_s} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})} \cdot \int_{\mathcal{D}} \mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{P}, \boldsymbol{D}, \boldsymbol{\phi}) \cdot \mathbb{P}(\boldsymbol{D} \mid \boldsymbol{A})\, d\boldsymbol{D}\, d\boldsymbol{P_s} \\
&\overset{\text{Eq.2.10}}{\approx} \int_{\mathcal{P}_s} \frac{\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\psi})}{\mathbb{P}(\boldsymbol{P_s}, \boldsymbol{\psi})} \cdot \mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\phi})\, d\boldsymbol{P_s}.
\end{aligned} \tag{2.15}$$

This is the basic model calculated several thousand times for different $\boldsymbol{\phi}$ by systematically sampling from the pools of plant information of the respective cores using MCMC techniques. In order to be able to describe this in more detail, certain framework conditions must first be introduced. To this end, we will now look at reference curves based on pollen and isotope information. As described in Sect. 2.1.1 and Sect. 2.1.2, many studies use such reference curves. These are shown, for example, in Fig. 2.2, Fig. 2.6, and the AP/NAP curve in Fig. 2.10. If a reconstruction according to Eq. 2.15 is performed for certain taxa weights $\boldsymbol{\omega}$, the expectation and/or median values of $\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta})$ can be compared with these reference curves. Here, the explained variances R^2 are used as a similarity measure and stored in a proxy pool $\boldsymbol{PP}$. Based on this idea, an extended BHM can be constructed (the weighting term is omitted for convenience):

$$\mathbb{P}(\boldsymbol{C}, \boldsymbol{\Theta} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{PP}) \propto \mathbb{P}(\boldsymbol{PP} \mid \boldsymbol{C}, \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{A}, \boldsymbol{\Theta}). \tag{2.16}$$

Note that the posterior distribution includes not only the climate variables as in Eq. 2.15, but also $\boldsymbol{\Theta}$. Finally, this model provides information on which taxa are important with respect to the selected reference curves. In this model, one could add a variety of additional reference curves based on, for

example, solar radiation or CO_2 (see Fig. 1.1). However, this work is limited to the consideration of proxies derived from isotopes and plant information. We know that some sections of the AP/NAP curves have fluctuations that are not due to climatological changes (e.g. Panagiotopoulos et al., 2013; Miebach et al., 2016; Schiebel, 2013; Neumann et al., 2007; Litt et al., 2012). In particular, the influence of humans on the vegetation around the lakes studied during the middle to late Holocene, complicates the interpretation of these curves. To account for such uncertainties, we specify a priori that the climate reconstructions should explain about 50 % of the variance of the respective reference curves. This offers on the one hand the possibility to follow the trends of the references and supports assumption 1 above that $\boldsymbol{P_s}$ provides sufficient variability. On the other hand, additional factors that influence the curves are taken into account. Thus, we specify the same prior distribution for each reference, namely beta distributions with a mean of 0.5 and a standard deviation of about 0.19. The latter leads to a better consideration of the wide range of lakes considered in this thesis. This results in a beta distribution with shape parameters 3. Assuming that all reference curves N_{proxy} are independent, the probability distributions of each component i of the vector $\boldsymbol{PP}$ can be defined as follows:

$$\mathbb{P}(PP_i \mid \cdot) = \text{Beta}(PP_i \mid 3, 3). \tag{2.17}$$

The parameters of the beta distributions can be changed individually depending on the lake studied and the reference proxy considered. However, for the purposes of this work, they represent a good compromise for the automatic calculation of all climate reconstructions for the entire European region, Northern Africa, and the Levant. At this point, instead of using multiple beta distributions, one might consider using the appropriate multivariate generalization. This is the Dirichlet distribution, which in our case has the disadvantage of being somewhat slower in MCMC sampling. In addition, other related metrics could be used in place of explained variance in future work. For example, the target redundance described in Glowienka-Hense et al., 2020 could be an alternative. This is the information that both reference curves provide equally, allowing them to be evaluated simultaneously.

Furthermore, we split the local reconstruction module as follows:

$$\mathbb{P}(\boldsymbol{C} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) = \mathbb{P}(\boldsymbol{C_p} \mid \boldsymbol{C_m}, \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{C_m} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}). \tag{2.18}$$

The first term on the right gives us the ability to constrain the reconstructions based on additional climate information from the past. These may be, for example, other local climate reconstructions from the same lake or from nearby lakes. In addition, prior distributions for specific time periods of the past could also be created from grid points of the particular lake from spatial reconstructions included in PMIP. It also allows reconstructions to be supplemented with information about past permafrost. Other information could come from experts who determine relatively accurate climate conditions for a given age based on vegetation studies of sediments. The last example of possible $\mathbb{P}(\boldsymbol{C_p} \mid \cdot)$ is presented and discussed in Sect. 3.3.5. The second term allows us to insert constraints

on the reconstructed modern climate. We define the transition from modern times to the past at 0 cal a BP, i.e. 1950 AD. This is because the temporal resolution of 50 years limits us, as we can only define the years 2000 AD or 1950 AD as the most recent period. However, since most of the cores of the EPD were recovered exactly in this interval, a compromise can thus be found for the definition of the youngest time interval. For modern climate, we have measurements from which we can use probability distributions as anchors for the reconstructions. More specifically, we use the Climate Research Unit (CRU) data from the Koeppen-Geiger classification rather than just one grid point from each site. This gives us a higher climatologically based variability that should simulate the climate of the catchment area of the lake or mire in question. We use the normal distribution as an approximation for the annual temperature and the gamma distribution for the annual precipitation. Again, it can be decided individually what to use as the basis for $\mathbb{P}(\boldsymbol{C_m} \mid \cdot)$, with our choice leading to a variety of useful climate reconstructions (see Sect. 3.4.1). All in all, we refer to the above as the prior climate module, which can be summarized as follows:

$$\mathbb{P}(\boldsymbol{C_p} \mid \boldsymbol{C_m}, \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) = \begin{cases} \mathbb{P}(\boldsymbol{C_p} \mid \cdot) & \text{if } \boldsymbol{C_p} \text{ is available,} \\ \text{Unif}(1, ..., \text{N}_{\text{age}}) & \text{otherwise,} \end{cases} \tag{2.19}$$

$$\mathbb{P}(\boldsymbol{C_m} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) = \begin{cases} \Gamma(\text{P}_{\text{ANN},m}) \text{ and } \mathcal{N}(\text{T}_{\text{ANN},m}) & \text{if } \boldsymbol{A}(C_1) \leq 0 \text{ cal BP,} \\ \text{Unif}(1, ..., \text{N}_{\text{age}}) & \text{otherwise.} \end{cases} \tag{2.20}$$

This means that reconstructions can be carried out with fewer restrictions even without prior climate information. This is made possible by the use of uniform distributions that encompass the reconstruction period and thus all time slices N_{age}. How useful such reconstructions ultimately are varies from case to case. Examples of this are discussed in Sect. 3.3.1 and Sect. 3.3.6.

Finally, we consider the third term of Eq. 2.16 in detail:

$$\mathbb{P}(\boldsymbol{P} \mid \boldsymbol{A}, \boldsymbol{\Theta}) = \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{A}, \boldsymbol{\phi}, \boldsymbol{\psi}) \approx \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{A}, \boldsymbol{\phi}) \approx \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi}). \tag{2.21}$$

First, we use assumption 3 above and split the parameter component. Then we state that $\boldsymbol{P}$ is independent of $\boldsymbol{\psi}$ when no $\boldsymbol{C}$ is given. Finally, the updated taxa weights $\mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi})$ are determined under the assumption that they are conditionally independent of A and thus hold for the entire reconstruction period. At this point, taxa weights could be split temporally based on additional prior information, so that they differ for specific time periods (e.g. glacials/interglacials). This approach is not explored further in this thesis and could be included in future work. The last term of Eq. 2.16, which is the joint distribution of $\boldsymbol{A}$ and $\boldsymbol{\phi}$, is treated as follows. First, we consistently follow the above reasoning and assume that all parameters $\boldsymbol{\Theta}$ are a priori independent of $\boldsymbol{A}$. Thus, this distribution can be formulated as follows:

$$\mathbb{P}(\boldsymbol{A}, \boldsymbol{\Theta}) = \mathbb{P}(\boldsymbol{A}) \cdot \mathbb{P}(\boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{\phi}). \tag{2.22}$$

Here, the last probability distribution describes the prior taxa weights, which are directly related to equation 2.21. The second term contains the parameters of the transfer functions and $\boldsymbol{A}$ is assumed to be uniform distributed if no depth or distance information are available. We see that already in the local reconstruction module in Eq. 2.15, the relations between $\boldsymbol{A}$ and $\boldsymbol{D}$ are inserted into our reconstruction scheme. With all the reformulations and simplifications listed above, Eq. 2.16 can be summarized as follows:

$$\begin{aligned}\mathbb{P}(\boldsymbol{C}, \boldsymbol{\Theta} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{PP}) \propto \mathbb{P}(\boldsymbol{PP} \mid \boldsymbol{C}, \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{C_p} \mid \boldsymbol{C_m}, \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \\ \cdot \mathbb{P}(\boldsymbol{C_m} \mid \boldsymbol{P}, \boldsymbol{A}, \boldsymbol{\Theta}) \cdot \mathbb{P}(\boldsymbol{\psi}) \cdot \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi}) \cdot \mathbb{P}(\boldsymbol{\phi}).\end{aligned} \tag{2.23}$$

Overall, the taxa should be weighted in such a way that those that better fit the constraints of the prior climate and proxy pool modules are weighted higher. How this is done in detail is described in the following. In the context of MCMC sampling, there are two ways to determine $\mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi})$. One of them is based on updating the weights through the random walk Metropolis-Hastings (rwMH). In the other case, an additional parameter allows the weights to be determined using Gibbs sampling. In the first case, $\boldsymbol{\phi}$ contains only the prior taxa weights $\boldsymbol{\omega}$. Without further prior information, we assume a uniform distribution across all taxa K at the beginning of the MCMC simulations:

$$\mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi}) = \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\omega}) = \text{Unif}(1, ..., \text{K}). \tag{2.24}$$

Such a Dirichlet distribution allows us to determine the taxa weights as we have requested above. This means that the taxa weights have values between 0 and 1 and add up to 1. The Jeffreys prior hyperparameters $\frac{1}{2}$ of this distribution give each taxon equal prior weight. Furthermore, these values provide a weaker constraint for determining the posterior taxa weights. This property follows directly from the characteristics of the Jeffreys prior (Gelman et al., 2013). Since a corresponding full conditional $\mathbb{P}(\boldsymbol{\omega} \mid \boldsymbol{P})$ does not follow a probability distribution from which we can sample directly, the taxa weights in this case are updated by a random walk Metropolis-Hastings.

In contrast, we proceed as follows with Gibbs sampling:

$$\mathbb{P}(\boldsymbol{P} \mid \boldsymbol{\phi}) = \mathbb{P}(\boldsymbol{P} \mid \boldsymbol{z}, \boldsymbol{\omega}) = \text{Unif}(1, ..., \text{K}). \tag{2.25}$$

The additionally introduced parameter $\boldsymbol{z}$ follows a categorical distribution, which leads to the following:

$$\mathbb{P}(\boldsymbol{z} \mid \boldsymbol{\omega}) = \text{Cat}(z_1, ..., z_\text{K} \mid \omega_1, ..., \omega_\text{K}). \tag{2.26}$$

This distribution requires additional information indicating how many taxa are drawn at once. We set this to t = K/2, which gives us the maximum number of distinct taxa combinations $\binom{\text{K}}{\text{t}}$. Here, too, the weights are ultimately determined by Eq. 2.30. Since the Dirichlet distribution is the conjugate prior to the categorical distribution, we can determine the full conditionals of $\boldsymbol{\omega}$ and $\boldsymbol{z}$. Specifically, $\boldsymbol{\omega}$

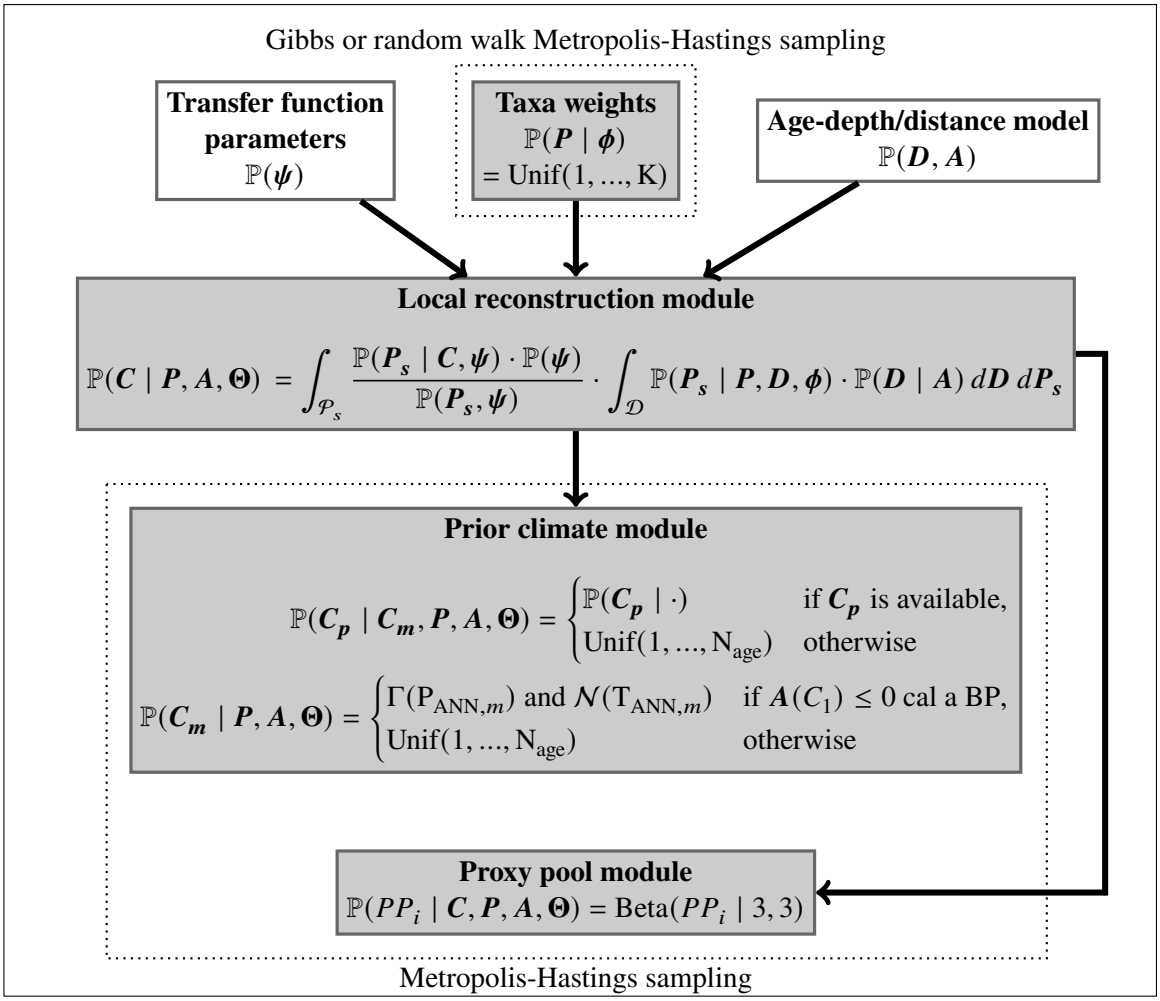


Figure 2.17: Directed acyclic graph of the Bayesian framework in Eq. 2.23. The gray boxes represent the quantities that will be inferred during MCMC sampling, and the white boxes contain fixed quantities. The corresponding arrows represent the mutual dependencies, with their direction pointing to the ascending hierarchy levels and the dashed boxes indicate the respective sampling procedures of the modules contained therein.

given z can be described as follows:

$$\mathbb{P}(\boldsymbol{\omega} \mid \boldsymbol{z}) = \text{Dir}(\frac{1}{2} + z_1, ..., \frac{1}{2} + z_{\text{K}}). \tag{2.27}$$

Moreover, we can determine the categorically distributed z given $\boldsymbol{\omega}$ and $\boldsymbol{P}$:

$$\mathbb{P}(\boldsymbol{z} \mid \boldsymbol{\omega}, \boldsymbol{P}) = \text{Cat}(\alpha_1, ...\alpha_{\text{K}}), \tag{2.28}$$

where, in our case, the following applies to each component α_i:

$$\alpha_{\mathrm{i}} \sim \omega_{\mathrm{i}}. \tag{2.29}$$

The respective weights are determined with the help of an additional prior distribution:

$$\mathbb{P}(\boldsymbol{\phi}) = \mathbb{P}(\boldsymbol{\omega}) = \mathrm{Dir}(\omega_1, ..., \omega_{\mathrm{K}} \mid \frac{1}{2}, ..., \frac{1}{2}). \tag{2.30}$$

Since we have determined a priori equally distributed weights, this relationship simplifies. However, if certain taxa are given a higher or lower weight at the beginning, this must be taken into account accordingly at this point.

Fig. 2.17 summarizes graphically how our local reconstruction framework works in general. The boxes in the first row contain the input variables, while the variables in the white boxes are not inferred during the MCMC simulation. For example, the parameters of the transfer functions are defined in Sect. 2.5. The middle upper gray box describing the inference of the taxa weights can be optionally determined via Gibbs or rwMH sampling. This is done by comparing the sampled climate reconstructions with other climate data and/or reference curves and constraining them accordingly. These comparisons are made using the Metropolis-Hastings sampling, represented by the lower dashed area.

2.5 Transfer functions via machine learning

This section describes the development of transfer functions. The main attention is paid to the probability distribution $\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi})$ from Eq. 2.15. We start with the idea of (Kühl et al., 2002), who uses multivariate Gaussian distributions for the presence of plants. For this purpose, analogous to this work, they took plant distribution maps as for the European olive tree in Fig. 2.18 (c) and assigned certain climate variables to them. In our case, these are the distributions of $\mathrm{P_{ANN}}$ and $\mathrm{T_{ANN}}$ over the years 1961 – 1990 AD from the CRU 4.01 dataset (Harris and Jones, 2017) and are shown in Fig 2.18 (a) and (b). These data, along with those from Fig. 2.18 (c), are presented in the climate feature space in Fig. 2.19. Sect. 2.1.1 already compiles the relationship between climate variability such as Heinrich, Bond, and DO events and isotopic information from speleothems. These fluctuations occurred within several decades to centuries and affected temperature and precipitation in regions of Eurasia and North America (Asmerom et al., 2010; Wagner et al., 2010; Genty, 2003; Benson et al., 1996). The assumed relationship between speleothems and these two climate variables differs from study to study. In the Levant, these are often used as proxies for the mean annual precipitation (Bar-Matthews et al., 2003; Cheng et al., 2015; Sánchez-Fernández et al., 2018; Bradley, 2014). However, in Central and Northern Europe it alternates between $\mathrm{T_{ANN}}$ and $\mathrm{P_{ANN}}$ (Mauri et al., 2015; Lauritzen and Lundberg, 1999). Since plants are also sensitive to such climatic variations (Gams,